RO2 - Lecture 3
3D-Rotation using Quaternions

we focus on students

Fachhochschule Dortmund
University of Applied Sciences and Arts

Recap

3D rotation matrices:

$$^{A}\boldsymbol{R}_{B} = \begin{bmatrix} r_{11} & r_{12} & r_{31} \\ r_{21} & r_{22} & r_{32} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

- Not very intuitive (low „human readability“)
- 9 elements required to describe a 3D rotation
- Compounding of rotations (by multiplications) computationally expensive

Euler Angles:

XYZ	XYZ	XZY	XZX
YXY	YXZ	YZX	YZY
ZXY	ZXZ	ZYX	ZYZ

- intuitive (high „human readability“)
- 3 elements required to describe a 3D rotation (still 9 for final rotation matrix)
- Main disadvantage: Gimbal lock condition

3D-Rotation using Quaternions

Definition of Quaternions

- Quaternions represent a number system which extends complex number (also hyper-complex number)
- Quaternions consist of a real element q_0 and three imaginary elements q_1, q_2, q_3

$$\boxed{\boldsymbol{q} = q_0 + i \cdot q_1 + j \cdot q_2 + k \cdot q_3}$$

- Common alternative notations are:

$$\boldsymbol{q} = q_w + i \cdot q_x + j \cdot q_y + k \cdot q_z$$

or

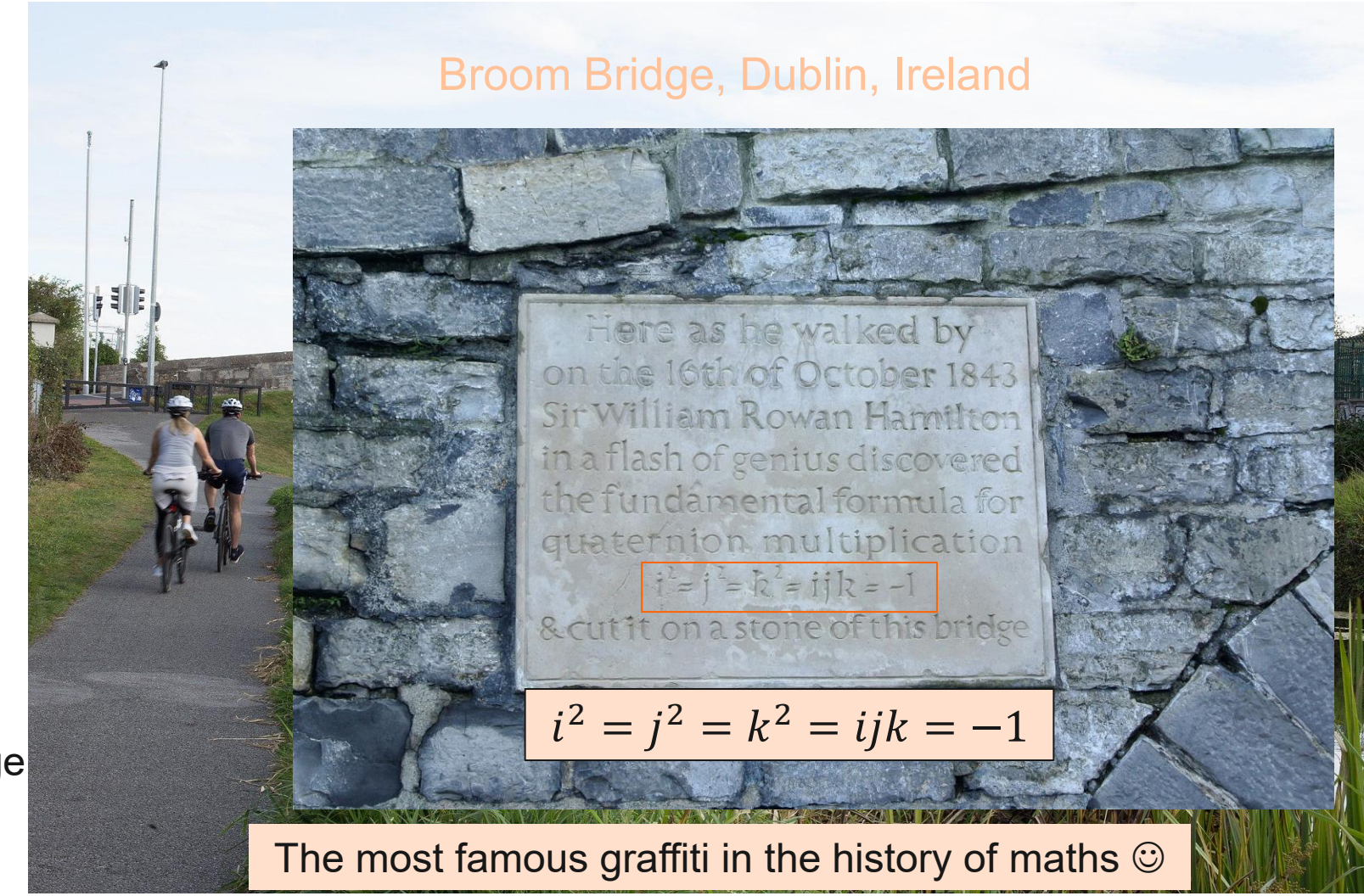
$$\boldsymbol{q} = s + i \cdot v_1 + j \cdot v_2 + k \cdot v_3$$

3D-Rotation using Quaternions

Quaternions were introduced by:

Sir William Rowan Hamilton
(1805–1865)

- Irish mathematician, astronomer, and physicist
- Professor of Astronomy at Trinity College Dublin
- Royal Astronomer of Ireland



3D-Rotation using Quaternions

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Annual event called the **Hamilton Walk** - scientists and mathematicians walk from Dunsink Observatory to the Royal Canal bridge in memory of Hamilton's discovery

Definition of Quaternions

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- Quaternions consist of a real element q_0 and three imaginary elements q_1, q_2, q_3

$$\boldsymbol{q} = q_0 + i \cdot q_1 + j \cdot q_2 + k \cdot q_3$$

- Vector form:

$$\boldsymbol{q} = [q_0 \quad q_1 \quad q_2 \quad q_3]$$

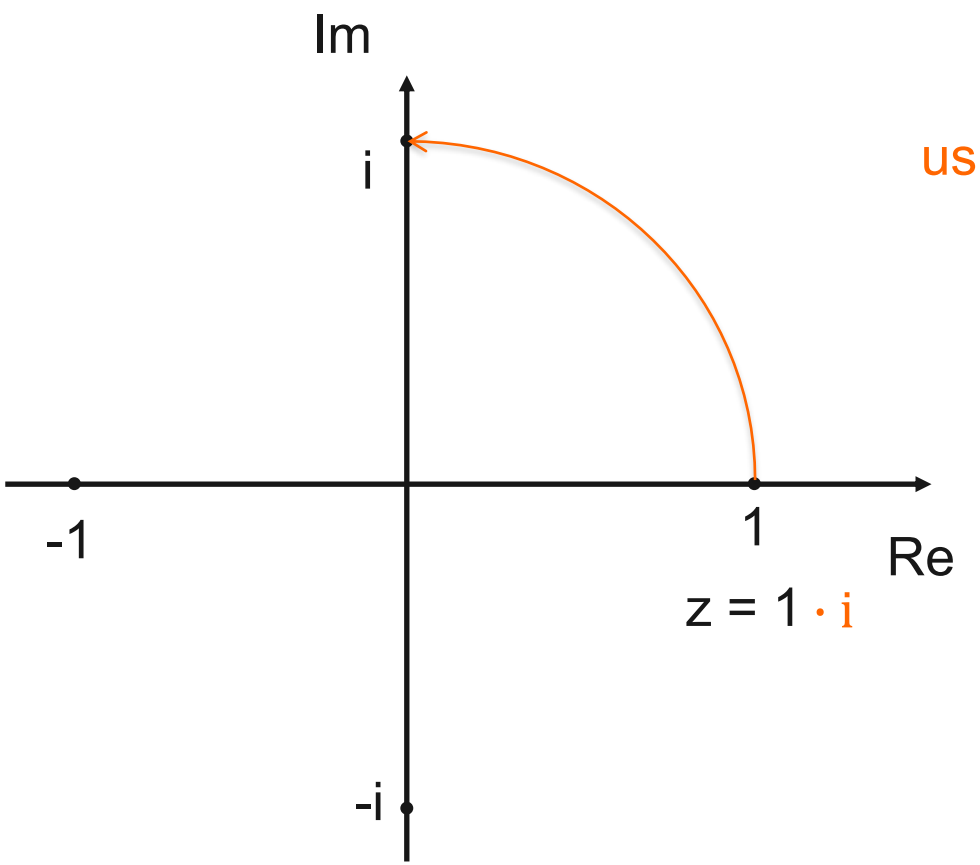
$$(q_0, \boldsymbol{v}), \quad \text{with} \quad \boldsymbol{v} = [q_1 \quad q_2 \quad q_3]$$

- i, j, k are defined such that

$$i^2 = j^2 = k^2 = ijk = -1$$

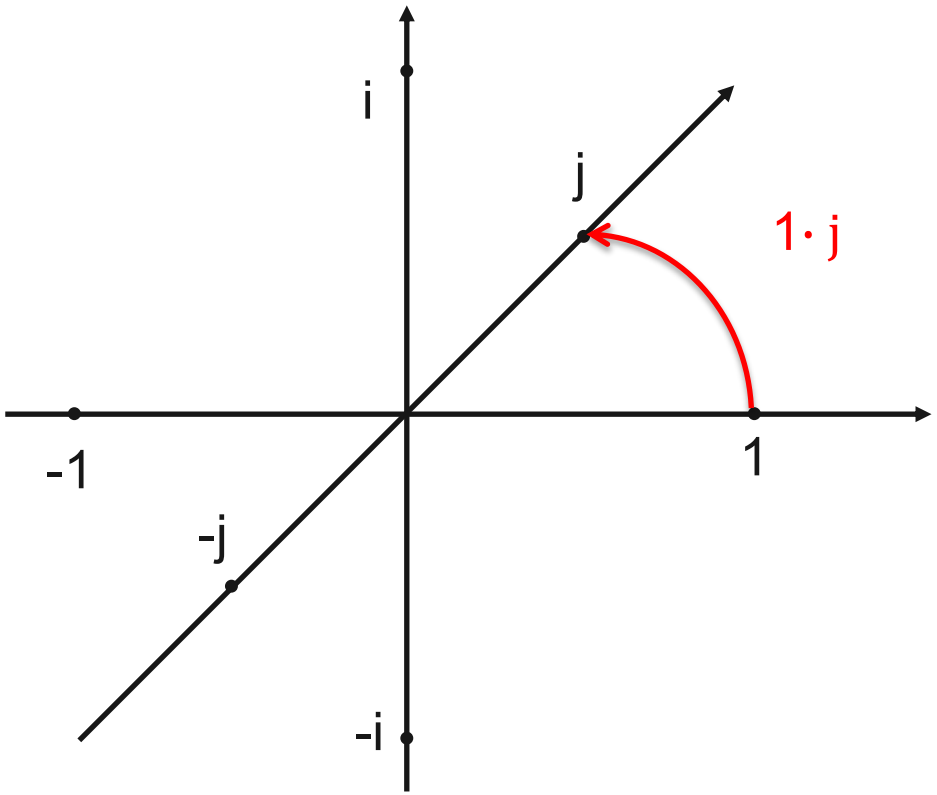
$$ij = k, \qquad jk = i, \qquad ki = j$$

Definition of Quaternions – Meaning of i, j, k (graphical)



usual complex space

Definition of Quaternions – Meaning of i, j, k (graphical)



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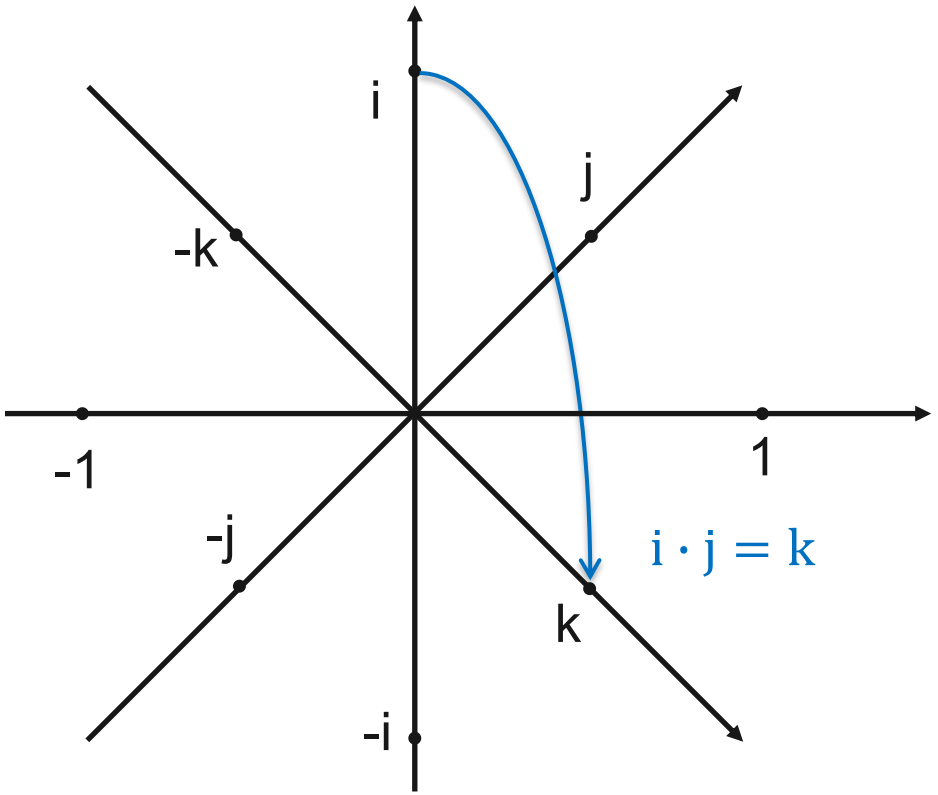


$$ij = k, \qquad jk = i, \qquad ki = j$$

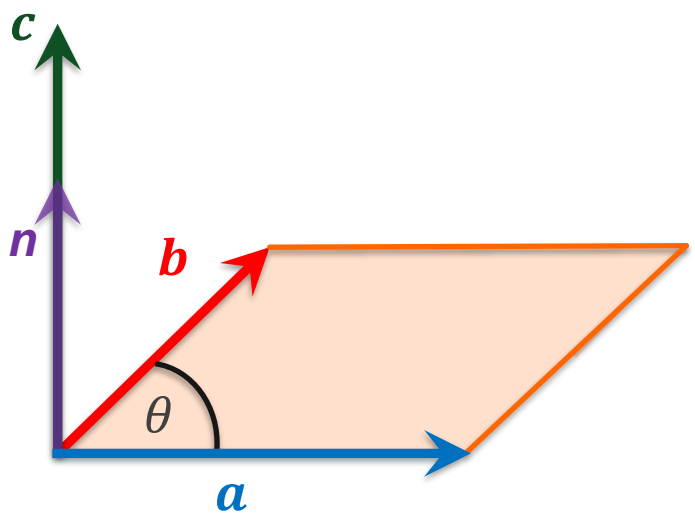
k is orthogonal to the *i*-*j*-space

The Product of two distinct imaginary quaternion units yields the third one → the axis orthogonal to the plane they span.

Definition of Quaternions – Meaning of i, j, k (graphical)

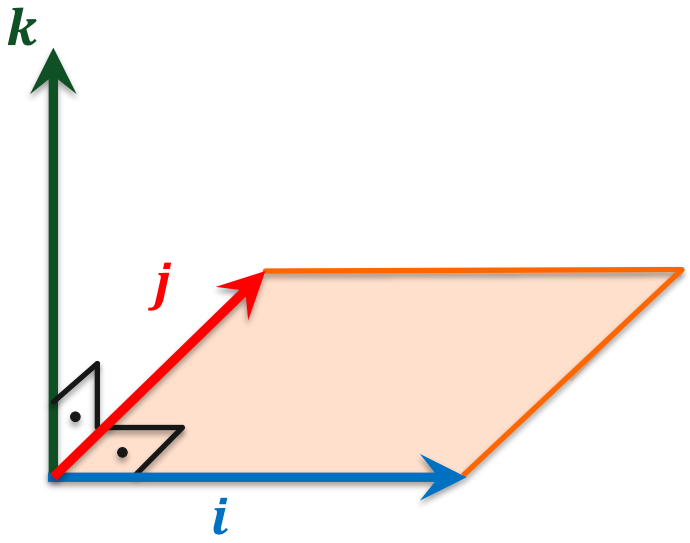
Similarity to cross product of unit/base vectors:

$$\boldsymbol{c} = \boldsymbol{a} \times \boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}|\sin(\theta)\boldsymbol{n} = \begin{bmatrix} \boldsymbol{a}_2\boldsymbol{b}_3 - \boldsymbol{a}_3\boldsymbol{b}_2 \\ \boldsymbol{a}_3\boldsymbol{b}_1 - \boldsymbol{a}_1\boldsymbol{b}_3 \\ \boldsymbol{a}_1\boldsymbol{b}_2 - \boldsymbol{a}_2\boldsymbol{b}_1 \end{bmatrix}$$



Definition of Quaternions – Meaning of i, j, k (graphical)

Similarity to cross product of unit/base vectors:



Consider a right handed coordinate system:

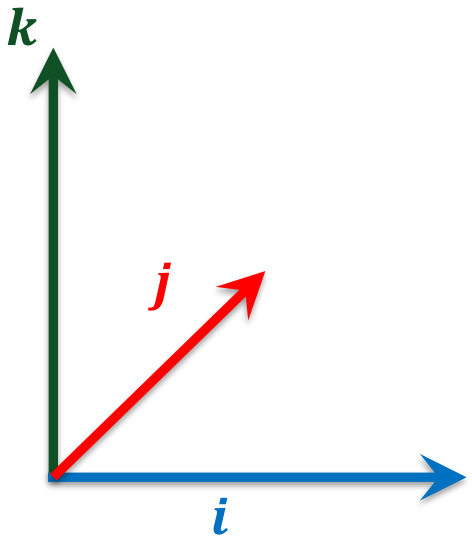
$$\boldsymbol{i} = e_x = [1 \quad 0 \quad 0]$$
$$\boldsymbol{j} = e_y = [0 \quad 1 \quad 0]$$
$$\boldsymbol{k} = e_z = [0 \quad 0 \quad 1]$$

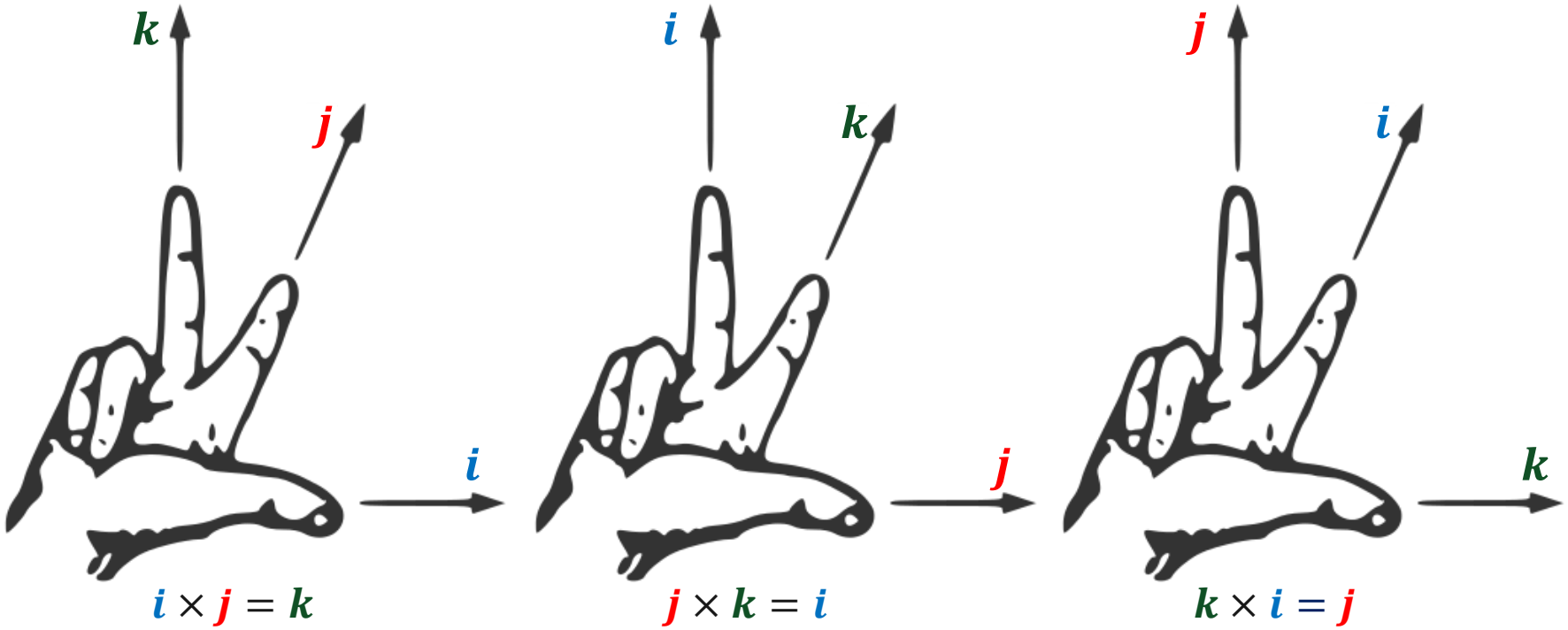
$$\boldsymbol{i} \times \boldsymbol{j} = \boldsymbol{k} \qquad \boldsymbol{j} \times \boldsymbol{i} = -\boldsymbol{k}$$
$$\boldsymbol{j} \times \boldsymbol{k} = \boldsymbol{i} \qquad \boldsymbol{k} \times \boldsymbol{j} = -\boldsymbol{i}$$
$$\boldsymbol{k} \times \boldsymbol{i} = \boldsymbol{j} \qquad \boldsymbol{i} \times \boldsymbol{k} = -\boldsymbol{j}$$

cross product unit vectors.mlx

Definition of Quaternions – Meaning of i, j, k (graphical)

Determining the vector orthogonal to the space span by the other two vectors:





Definition of Quaternions – Meaning of i, j, k (graphical)

Difference between quaternion algebra and cross products of unit vectors

Cross products of unit vectors:

$$i \times j = k, \qquad j \times k = i, \qquad k \times i = j$$

Products of quaternion components:

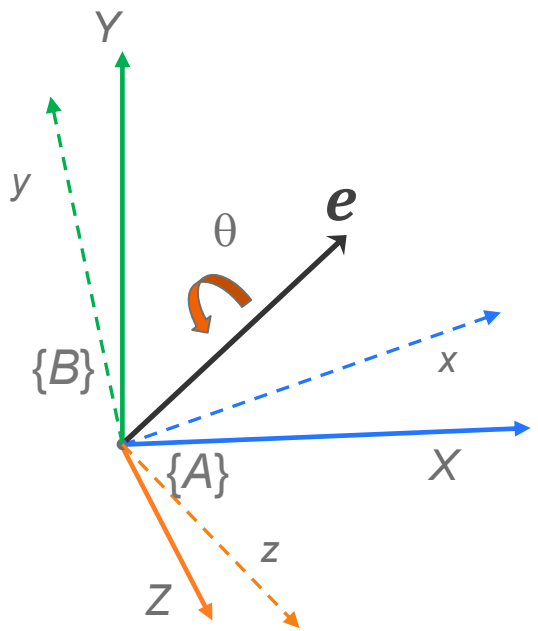
$$ij = k, \qquad jk = i, \qquad ki = j$$

$$i^2 = j^2 = k^2 = ijk = -1$$

- In vector algebra we use the cross product $\times$
- In quaternion algebra we use quaternion multiplication as defined by Hamilton's rules
- In both cases, the result is an axis orthogonal to the plane

Unit Quaternions for representation of 3D rotations

- For 3D rotations → use unit quaternions: $|\boldsymbol{q}| = \sqrt{{q_0}^2 + {q_1}^2 + {q_2}^2 + {q_3}^2} = 1$
- Scalar part q_0 represents degree of rotation, vector part $\boldsymbol{v} = [q_1 \quad q_2 \quad q_3]$ the rotation axis



If we have a rotation axis defined as unit vector $\boldsymbol{e} = [e_x \quad e_y \quad e_z]^T$ and θ is the rotation angle about that axis, then:

$$q_0 = \cos\left(\frac{1}{2}\theta\right)$$

$$q_1 = e_x \cdot \sin\left(\frac{1}{2}\theta\right), q_2 = e_y \cdot \sin\left(\frac{1}{2}\theta\right), q_3 = e_z \cdot \sin\left(\frac{1}{2}\theta\right)$$

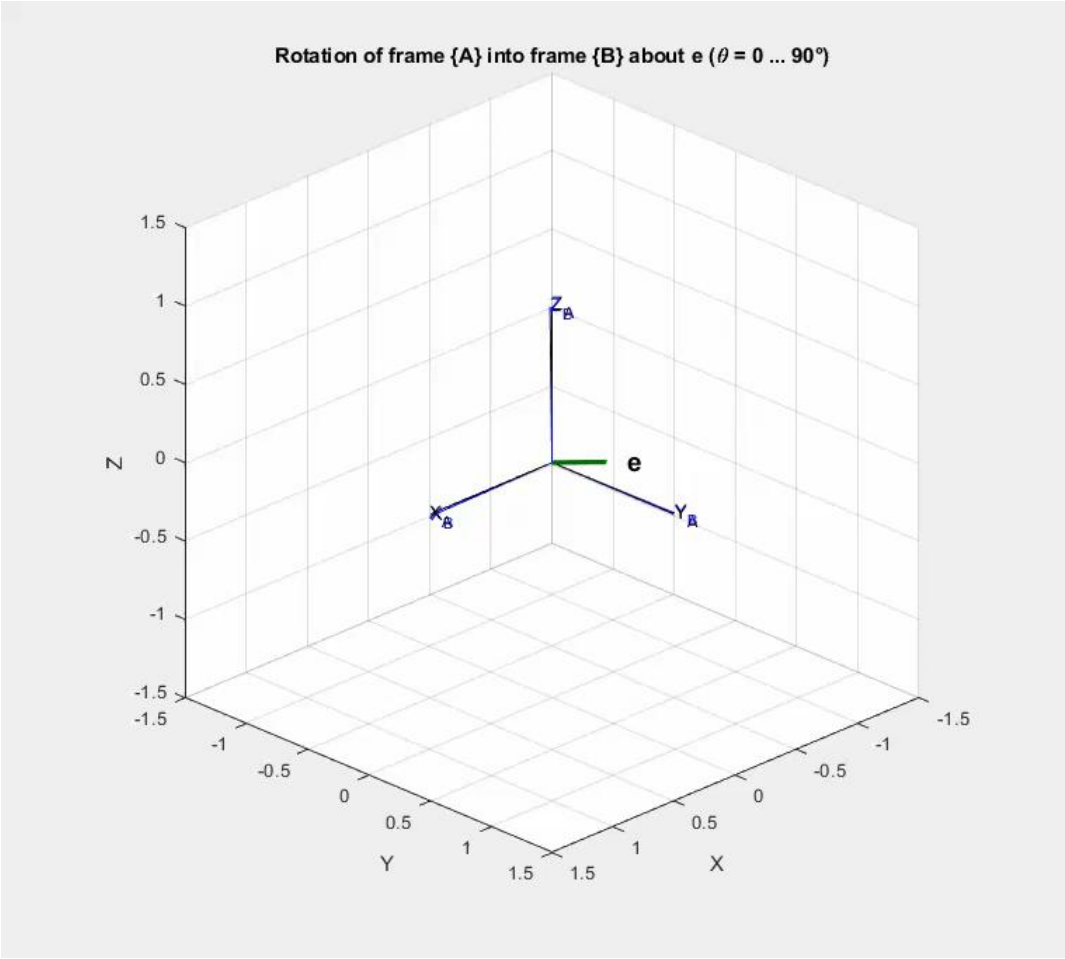
$$\Rightarrow \boldsymbol{q} = \left[\cos\left(\frac{1}{2}\theta\right), \boldsymbol{e} \cdot \sin\left(\frac{1}{2}\theta\right)\right]$$

$\boldsymbol{q}$ represents the orientation of local frame *{B}* with axes *x*, *y*, *z* in relation to reference frame *{A}* with axes *X*, *Y*, *Z*

$$\boldsymbol{q} = {}^{A}\boldsymbol{q}_{\boldsymbol{B}}$$

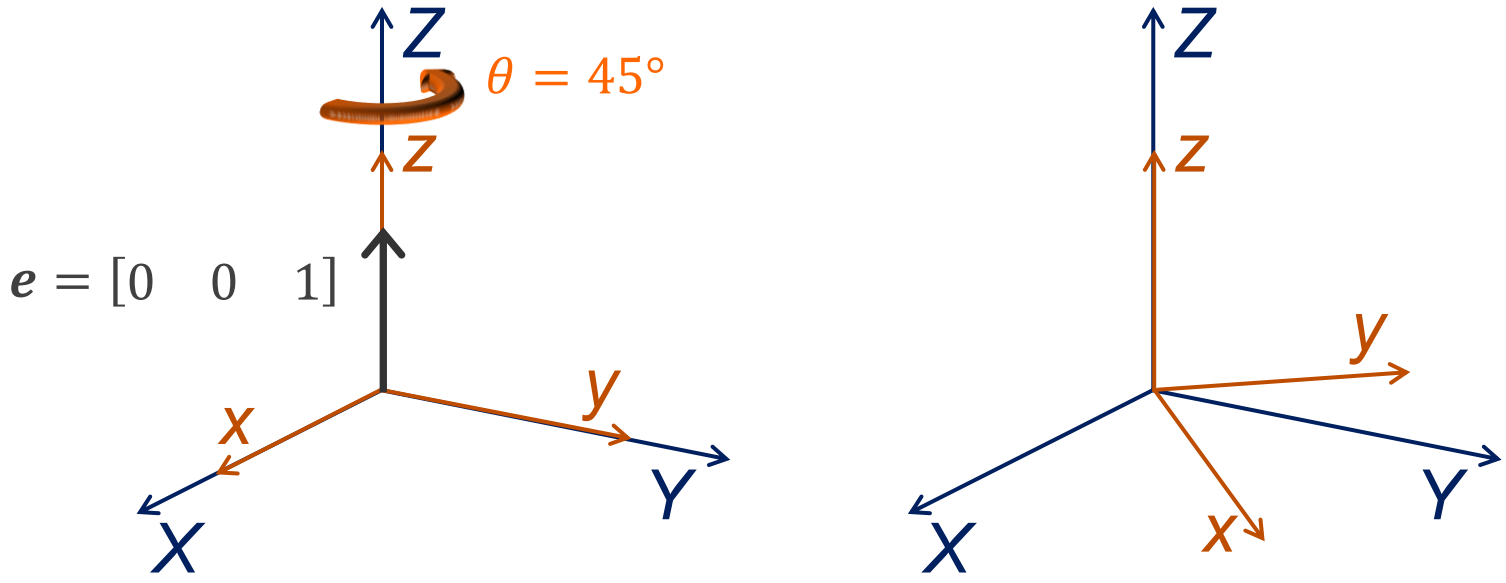
Unit Quaternions for representation of 3D rotations

Animated example with e = $[0.41 \quad 0.82 \quad 0.41]$, $\theta = 0 \ldots 90°$



Unit Quaternions for representation of 3D rotations

- E.g.: rotation of 45° regarding *z*-axis



$$q_0 = \cos\left(\frac{1}{2}\theta\right) = \cos\left(\frac{1}{2}\pi/4\right)$$

$$q_1 = e_x \cdot \sin\left(\frac{1}{2}\theta\right), q_2 = e_y \cdot \sin\left(\frac{1}{2}\theta\right), q_3 = e_z \cdot \sin\left(\frac{1}{2}\theta\right) \Rightarrow q_1 = 0, q_2 = 0, q_3 = 1 \cdot \sin\left(\frac{1}{2}\pi/4\right)$$

$$\Rightarrow \boldsymbol{q} = \left[\cos\left(\frac{1}{2}\theta\right), \vec{\boldsymbol{e}} \cdot \sin\left(\frac{1}{2}\theta\right)\right] = \left[\cos\left(\frac{1}{2}\pi/4\right), [0 \quad 0 \quad 1] \cdot \sin\left(\frac{1}{2}\pi/4\right)\right]$$

3D-Rotation using Quaternions

Unit Quaternions for representation of 3D rotations

- Inverse $\boldsymbol{q}^{-1}$ of a unit quaternion is given by its‘ conjugation $\boldsymbol{q}^*$

$$\boldsymbol{q}^{-1} = \boldsymbol{q}^* = [q_0 \quad -q_1 \quad -q_2 \quad -q_3]$$

- Inverse operation inverts the rotation direction

$$\left({}^{A}\boldsymbol{q}_{B}\right)^{-1} = {}^{B}\boldsymbol{q}_{A}$$

- Similar to rotation matrices: compounding of rotation by multiplication, e.g.

$$\boldsymbol{q}_{\mathrm{ab}} = \boldsymbol{q}_a \circ \boldsymbol{q}_{\mathrm{b}}$$

- Multiplication of two quaternions $\boldsymbol{a} \circ \boldsymbol{b} = [a_0 \quad a_1 \quad a_2 \quad a_3] \circ [b_0 \quad b_1 \quad b_2 \quad b_3]$

$$= \begin{bmatrix} a_0b_0 - a_1b_1 - a_2b_2 - a_3b_3 \\ a_0b_1 + a_1b_0 + a_2b_3 - a_3b_2 \\ a_0b_2 - a_1b_3 + a_2b_0 + a_3b_1 \\ a_0b_3 + a_1b_2 - a_2b_1 + a_3b_0 \end{bmatrix}^T$$

Non commutative:
$a \cdot b \neq b \cdot a$

Quaternion Multiplication (Hamilton Product)

$$\boldsymbol{a} \circ \boldsymbol{b} = (a_0 + a_1 \cdot i + a_2 \cdot j + a_3 \cdot k) \cdot (b_0 + b_1 \cdot i + b_2 \cdot j + b_3 \cdot k)$$

$$= \begin{array}{c} a_0 b_0 + a_0 b_1 i + a_0 \mathrm{b}_2 j + a_0 b_3 k \\ + a_1 b_0 i + a_1 b_1 i^2 + \mathrm{a}_1 b_2 ij + \mathrm{a}_1 b_3 ik \\ + a_2 b_0 j + a_2 b_1 ji + a_2 b_2 j^2 + a_2 b_3 jk \\ + a_3 b_0 k + a_3 b_1 ki + a_3 b_2 kj + a_3 b_3 k^2 \end{array}$$

$$= \begin{array}{c} a_0 b_0 + a_0 b_1 i + a_0 \mathrm{b}_2 j + a_0 b_3 k \\ + a_1 \mathrm{b}_0 i - a_1 b_1 + \mathrm{a}_1 b_2 \mathrm{k} - \mathrm{a}_1 b_3 j \\ + a_2 b_0 j - a_2 b_1 k - a_2 b_2 + a_2 b_3 i \\ + a_3 b_0 k + a_3 b_1 j - a_3 b_2 i - a_3 b_3 \end{array}$$

$$= \begin{array}{c} a_0 b_0 - a_1 b_1 - a_2 b_2 - a_3 b_3 \\ + i(a_0 b_1 + a_1 \mathrm{b}_0 + a_2 b_3 - a_3 b_2) \\ + j(a_0 \mathrm{b}_2 - \mathrm{a}_1 b_3 + a_2 b_0 + a_3 b_1) \\ k(a_0 b_3 + \mathrm{a}_1 b_2 - a_2 b_1 + a_3\, b_0) \end{array} \qquad = \begin{bmatrix} a_0 b_0 - a_1 b_1 - a_2 b_2 - a_3 b_3 \\ a_0 b_1 + a_1 \mathrm{b}_0 + a_2 b_3 - a_3 b_3 \\ a_0 \mathrm{b}_2 - \mathrm{a}_1 b_3 + a_2 b_0 + a_3 b_1 \\ a_0 b_3 + \mathrm{a}_1 b_2 - a_2 b_1 + a_3\, b_0 \end{bmatrix}^T$$

·	1	i	j	k
1	1	i	j	k
i	i	-1	k	-j
j	j	-k	-1	i
k	k	j	-i	-1

Unit Quaternions for representation of 3D rotations

- Example for compounding of orientations ${}^{A}\boldsymbol{q}_B$ and ${}^{B}\boldsymbol{q}_C$ to get compounded orientation ${}^{A}\boldsymbol{q}_C$:

$$ {}^{A}\boldsymbol{q}_C = {}^{A}\boldsymbol{q}_B \circ {}^{B}\boldsymbol{q}_C $$

- Example for using inverse to revert a rotation ${}^{B}\boldsymbol{q}_C$:

$$ {}^{A}\boldsymbol{q}_C \circ {}^{B}\boldsymbol{q}_C{}^{-1} = {}^{A}\boldsymbol{q}_C \circ {}^{C}\boldsymbol{q}_B = {}^{A}\boldsymbol{q}_B $$

- "Identity" quaternion (no rotation) defined as:

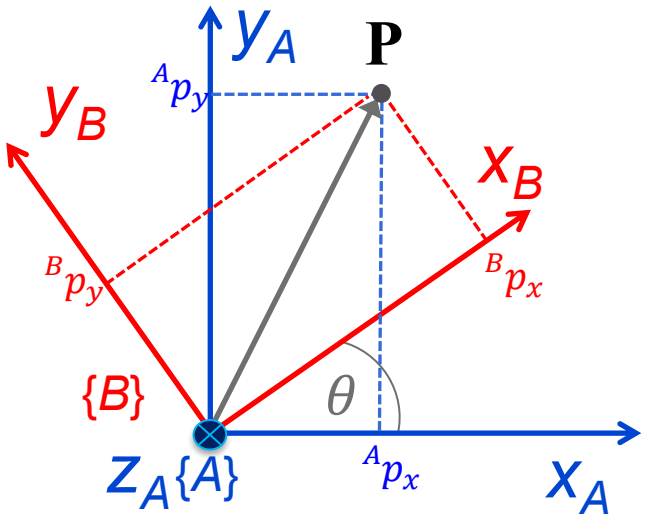
$$ \boldsymbol{q} = [1 \quad 0 \quad 0 \quad 0] $$

$\boldsymbol{a} \circ \boldsymbol{b}$ with $\boldsymbol{a} = [1 \quad 0 \quad 0 \quad 0]$

$$ \boldsymbol{a} \circ \boldsymbol{b} = \begin{bmatrix} a_0 b_0 - a_1 b_1 - a_2 b_2 - a_3 b_3 \\ a_0 b_1 + a_1 b_0 + a_2 b_3 - a_3 b_2 \\ a_0 b_2 - a_1 b_3 + a_2 b_0 + a_3 b_1 \\ a_0 b_3 + a_1 b_2 - a_2 b_1 + a_3 b_0 \end{bmatrix}^T = \begin{bmatrix} 1 b_0 - 0 b_1 - 0 b_2 - 0 b_3 \\ 1 b_1 + 0 b_0 + 0 b_3 - 0 b_2 \\ 1 b_2 - 0 b_3 + 0 b_0 + 0 b_1 \\ 1 b_3 + 0 b_2 - 0 b_1 + 0 b_0 \end{bmatrix}^T = [b_0 \quad b_1 \quad b_2 \quad b_3] = \boldsymbol{b} $$

Describing a vector before and after rotation

Recap from rotation matrices…

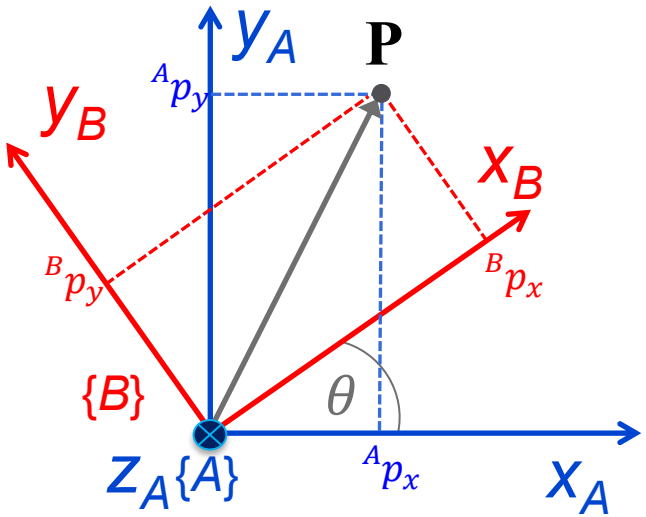


$$^A\boldsymbol{p} = \begin{bmatrix} ^Ap_x \\ ^Ap_y \\ ^Ap_z \end{bmatrix} \qquad ^B\boldsymbol{p} = \begin{bmatrix} ^Bp_x \\ ^Bp_y \\ ^Bp_z \end{bmatrix}$$

$$^A\boldsymbol{p} = {}^A\boldsymbol{R}_B \; {}^B\boldsymbol{p}$$

$$^B\boldsymbol{p} = {}^A\boldsymbol{R}_B^{-1} \; {}^A\boldsymbol{p} = {}^B\boldsymbol{R}_A \; {}^A\boldsymbol{p}$$

Describing a vector before and after rotation



$$^A\boldsymbol{p} = \begin{bmatrix} ^Ap_x \\ ^Ap_y \\ ^Ap_z \end{bmatrix} \qquad ^B\boldsymbol{p} = \begin{bmatrix} ^Bp_x \\ ^Bp_y \\ ^Bp_z \end{bmatrix}$$

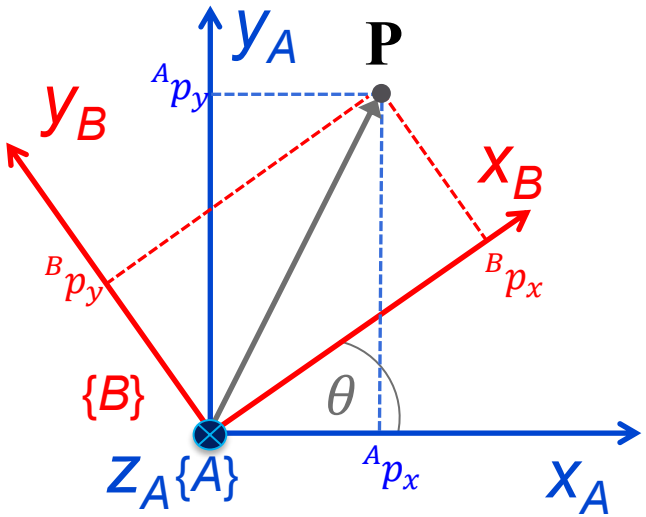
With quaternions:

1. make quaternion from vector $\boldsymbol{p}$: $^B\boldsymbol{q} = [0 \quad ^B\boldsymbol{p}]$
2. Rotate by quaternion $^A\boldsymbol{q}_B$:

$$^A\boldsymbol{q} = {}^A\boldsymbol{q}_B \; {}^B\boldsymbol{q} \; {}^A\boldsymbol{q}_B^* = [0 \quad ^A\boldsymbol{p}]$$

$$^B\boldsymbol{q} = {}^B\boldsymbol{q}_A \; {}^A\boldsymbol{q} \; {}^B\boldsymbol{q}_A^* = [0 \quad ^B\boldsymbol{p}]$$

Describing a vector before and after rotation



$$^A\boldsymbol{p} = \begin{bmatrix} ^Ap_x \\ ^Ap_y \\ ^Ap_z \end{bmatrix} \qquad ^B\boldsymbol{p} = \begin{bmatrix} ^Bp_x \\ ^Bp_y \\ ^Bp_z \end{bmatrix}$$

With quaternions:

Alternatively: $^A\boldsymbol{q}_B \rightarrow {}^A\boldsymbol{R}_B$

$$^A\boldsymbol{p} = {}^A\boldsymbol{R}_B \; {}^B\boldsymbol{p}$$

$$^B\boldsymbol{p} = {}^A\boldsymbol{R}_B^{-1} \; {}^A\boldsymbol{p} = {}^B\boldsymbol{R}_A \; {}^A\boldsymbol{p}$$

3D-Rotation using Quaternions

Conversion of Unit Quaternions to other Rotation Representations:

- Conversion of quaternion ${}^{A}\boldsymbol{q}_B$ to a rotation matrix ${}^{A}\boldsymbol{R}_B$ (for readability: $\boldsymbol{q} = {}^{A}\boldsymbol{q}_B$) [1]:

$$
{}^{A}\boldsymbol{R}_B = \begin{bmatrix} 2{q_0}^2 - 1 + 2{q_1}^2 & 2(q_1 q_2 + q_0 q_3) & 2(q_1 q_3 + q_0 q_2) \\ 2(q_1 q_2 - q_0 q_3) & 2{q_0}^2 - 1 + 2{q_2}^2 & 2(q_2 q_3 + q_0 q_1) \\ 2(q_1 q_3 + q_0 q_2) & 2(q_2 q_3 + q_0 q_1) & 2{q_0}^2 - 1 + 2{q_3}^2 \end{bmatrix}
$$

Conversion of Unit Quaternions to other Rotation Representations:

- Conversion of quaternion ${}^{A}\boldsymbol{q}_B$ to euler angles for sequence *ZYX* [2]:

$$\psi = \operatorname{atan2}\left(2(q_2 q_3 + q_0 q_1), 1 - 2({q_0}^2 + {q_3}^2)\right)$$
$$\theta = -\arctan\left(\frac{2(q_1 q_3 + q_0 q_2)}{\sqrt{1 - (2q_1 q_3 + 2q_0 q_2)^2}}\right)$$
$$\varphi = \operatorname{atan2}(2(q_1 q_2 + q_0 q_3), 2{q_0}^2 - 1 + 2{q_1}^2)$$

- Conversion of euler angles (*ZYX*) to quaternion ${}^{A}\boldsymbol{q}_B$ [5]:

$$q_0 = \cos\left(\frac{\varphi}{2}\right)\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\psi}{2}\right) + \sin\left(\frac{\varphi}{2}\right)\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{\psi}{2}\right)$$
$$q_1 = \sin\left(\frac{\varphi}{2}\right)\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\psi}{2}\right) - \cos\left(\frac{\varphi}{2}\right)\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{\psi}{2}\right)$$
$$q_2 = \cos\left(\frac{\varphi}{2}\right)\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\psi}{2}\right) + \sin\left(\frac{\varphi}{2}\right)\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{\psi}{2}\right)$$
$$q_3 = \cos\left(\frac{\varphi}{2}\right)\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{\psi}{2}\right) - \sin\left(\frac{\varphi}{2}\right)\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\psi}{2}\right)$$

3D-Rotation using Quaternions

Advantages of Quaternions:

- No singularities / no gimbal lock effect
- 3D rotation in a single representation → no convention or rotation sequence to be considered
- Easy and „smooth“ interpolation between orientations (see SLERP- und SQUAD-algorithms)
- Combination of rotations very efficient:
  - Multiplication of two 3x3 rotation matrices:
    27 multiplications, 18 additions
  - Multiplication of two quaternions:
    16 multiplications, 12 additions

Disadvantage of Quaternions:

- Less intuitive/human readable than Euler Angles

3D-Rotation using Quaternions

RTB functions

	function	description	Comment
Constructors	***q*** = *UnitQuaternion*	Create a unit quaternion corresponding to no rotation (similar to identity matrix)	
	q = *UnitQuaternion*(q_0, $\boldsymbol{v}$)	Create a unit quaternion where q_0 is the scalar element, $\boldsymbol{v}$ the 3x1 vector part	
	q = *UnitQuaternion*($\boldsymbol{R}$)	Create a unit quaternion which corresponds to the orientation given by the 3x3 rotation matrix ***R***	
	q = *UnitQuaternion.rpy(roll, pitch, yaw)*	Create a unit quaternion which corresponds to the roll, pitch and yaw values (by default, these correspond to a rotation about Z, Y, X; other conventions via options)	By default: angles in radians
Functions	***q***.*plot()*	Plot the the unit quaternion as an oriented coordinate frame (similar to trplot)	
	q.*animate*($\boldsymbol{q}_f$)	Animates a 3D coordinate frame moving from orientation ***q*** to orientation $\boldsymbol{q}_f$.	
	rpy = ***q***.*torpy()*	Convert quaternion into roll, pitch and yaw angles (ZYX by default)	By default: angles in radians returned
	q.*inv*	Generate the inverse of a quaternion (alternativly: inv(***q***))	
	$\boldsymbol{q}_1 * \boldsymbol{q}_2$	Quaternion (Hamilton) product of $\boldsymbol{q}_1$ and $\boldsymbol{q}_2$	

Exercises… see ILIAS ;-)

Sources

[1] J. B. Kuipers. *Quaternions and Rotation Sequences: A Primer with Applications to Orbits, Aerospace and Virtual Reality*. Princeton University Press, 1999.

[2] Mei Wang, Yunchun Yang, R.R. Hatch, and Yanhua Zhang. Adaptive Iter for a miniature mems based attitude and heading reference system. *Position Location and Navigation Symposium, 2004*. PLANS *2004*, pages 193 - 200, apr. 2004.

[3] Itzhack Y Bar-Itzhack. New method for extracting the quaternion from a rotation matrix. *AIAA Journal of Guidance, Control and Dynamics*, 23(6):10851087, Nov.Dec 2000.

[4] Sarabandi, Soheil & Thomas, Federico. (2018). A Survey on the Computation of Quaternions From Rotation Matrices. *Journal of Mechanisms and Robotics*. 11. 10.1115/1.4041889.

[5] NASA Mission Planning and Analysis Division (July 1977). "*Euler Angles, Quaternions, and Transformation Matrices*". NASA. Retrieved 24 May 2021.